Halcyon Aerospace Ltd.

Project KESTREL: Annual Technical Review 2025

Executive Summary

This report reviews the progress of Project KESTREL, Halcyon Aerospace's autonomous medical-supply delivery drone program, during the 2025 calendar year. The program is led by Dr. Meera Iyer, Chief Engineer, and is headquartered in Pune, Maharashtra.

The current production candidate, the KESTREL-7, completed its full certification test campaign this year. It carries medical payloads over a range of more than 60 km and has demonstrated a flight reliability that exceeds the program's original target of 97%.

The total approved budget for the program is Rs. 48 crore. Key risks identified this year relate to weather tolerance and cold-temperature battery performance, both discussed later in this review.

Program Background and Team

Project KESTREL was founded in 2021 to address delays in delivering vaccines, blood units and emergency medicines to rural primary health centres in Karnataka and Maharashtra.

Dr. Meera Iyer joined Halcyon in 2019 after completing her PhD in control systems at IIT Madras. Before leading KESTREL, she worked on flight-control software for agricultural spraying drones.

The program has two external research partners. The Indian Institute of Science, Bengaluru, contributes navigation and sensor-fusion research. Nordwind Batteries GmbH, based in Stuttgart, Germany, supplies the solid-state battery cells used in the KESTREL-7.

All outdoor flight testing takes place at the Halcyon test range near Chitradurga, Karnataka. The core engineering team consists of 34 engineers, of whom 11 work on software and autonomy.

Technical Specifications of KESTREL-7

The table below lists the certified specifications of the KESTREL-7 as of December 2025.

Parameter	Value
Maximum payload	4.5 kg
Maximum range	62 km
Cruise speed	88 km/h
Endurance (flight time)	71 minutes
Empty weight	9.2 kg
Battery	12,800 mAh solid-state (Nordwind)
Navigation	Dual GNSS with visual odometry backup
Firmware version (certified)	v3.2.4

Payloads are carried in a temperature-controlled pod that keeps contents between 2 and 8 degrees Celsius, which is required for most vaccines.

Flight Testing Results

During 2025 the team conducted 1,340 test flights. Of these, 17 flights ended in a failure, defined as any flight that did not deliver its payload to the intended destination.

Root-cause analysis attributed the failures as follows: 9 were caused by GPS drift in areas with poor satellite coverage, 5 were caused by motor overheating during long climbs, and 3 were caused by software faults.

This corresponds to a mission success rate of 98.7%, exceeding the 97% target. After the dual-GNSS upgrade in August, no further GPS-drift failures were recorded.

Incident Report: Chitradurga, 14 March 2025

On 14 March 2025, the prototype KESTREL-6B lost altitude control shortly after take-off and crashed in an unoccupied field near the Chitradurga test range. No one was injured and no third-party property was damaged.

The investigation traced the crash to a bug in firmware version v3.2.1, where an integer overflow in the altitude-hold routine produced an incorrect climb command. The bug was fixed in firmware v3.2.4, which is now the certified version on all KESTREL-7 aircraft.

Following the incident, all test flights were paused for 12 days while the fix was validated in simulation.

Known Limitations

The KESTREL-7 is not cleared to fly in rainfall heavier than 8 mm per hour, or in sustained winds above 38 km/h. Under these conditions operations are automatically grounded by the mission planner.

Cold temperatures also affect performance. Below 5 degrees Celsius, battery output drops and the effective range falls by approximately 30%. This mainly affects winter operations in hilly districts.

The aircraft cannot currently operate at night, pending regulatory approval for beyond-visual-line-of-sight night flights.

Budget and Future Plans

Of the approved Rs. 48 crore budget, Rs. 41.5 crore had been spent by the end of 2025, primarily on prototype manufacturing, battery procurement and flight testing.

The next-generation KESTREL-8 is planned for first flight in Q3 2026. Its design target is a maximum payload of 7 kg and improved cold-weather battery management.

A commercial pilot program will begin in Hyderabad in partnership with the pharmacy chain MedRoute, delivering prescription medicines to residential areas.

Conclusion

Project KESTREL met its 2025 reliability goals, resolved its most serious safety incident, and is on track for commercial deployment. Weather tolerance and cold-weather range remain the priority engineering challenges for 2026.